* CI-CD-pipeline project *

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Must Know

1 Docker

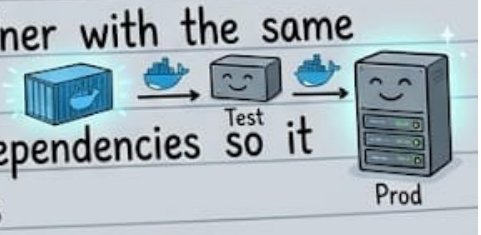
purpose : Package the application with everything it needs to run

Problem before Docker :

"Works on my machine, But not on Server"

- Different machines have node
- different Node versions
- missing libraries
- different environments

Solution : Docker creates a container with the same environment everywhere Means:



docker packages the app + dependencies so it runs the same on every machines

Benefits:
- ✓ No environment problems
- ✓ Easy deployment
- ✓ Same setup for dev, test & production
- ✓ Standardize environment

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with

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Must Know : GitHub Actions

purpose: → Automatically runs tasks when code changes

Problem before CI/CD

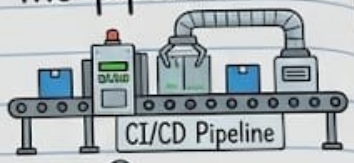
manually do

- ran tests
- build app
- build docker image
- deploy

This is slow and error-prone

Solution: **Githubaction / Jenkins** automates the pipeline

Example workflow :



Developer pushes code

↓

Tests run automatically

↓

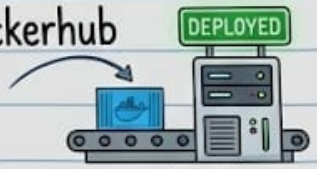
Docker image builds

↓

Image pushed to Dockerhub

↓

Deployment starts



GithubActions automates testing, building & deployment when code is pushed.

Benefits : ✓ Automation

ping ✓ Faster development

ping ✓ Fewer human errors

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✳ Simple Real Workflow with Learn with Sameer Tiruwa

① Must Know : Kubernetes (K8s)

✓ Developer pushes code

Github Actions runs pipeline

↓ Docker image created

Pipeline

↓ Image sent to Dockerhub

Virtual warehouse

Github Actions automates testing, building & deployment when code is pushed.

purpose: → Scale and Manage Container

Automatically runs tasks when code changes

Problem before Kubernetes:

Running container manually is hard when you have:
- many container,
- traffic spikes,
- container crashes

This is slow and error-prone

Solution: Kubernetes manages container automatically

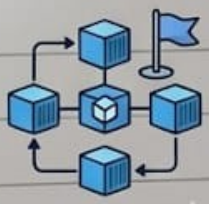
Ovese things are fixed automatically

App crashes → Kubernetes restarts it

Traffic increases → Kubernetes scales pods

Kubernetes manages and scales Docker container automatically

Benefits:



✓ Automation ✓ Self Healing ✓ Load Balancing ✓ Production-ready systems